

MATH 551: NUMERICAL ORDINARY DIFFERENTIAL EQUATIONS

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1. AUGUST 24, 2026

1.1. **Ordinary differential equations.** An ODE has the form

$$x'(t) = f(t, x(t)).$$

Given an initial condition $x(t_0) = x_0$, the corresponding *initial-value problem* (IVP) is to find $x(t)$ for $t \geq t_0$ such that

$$(1) \quad x'(t) = f(t, x(t)), \quad x(t_0) = x_0.$$

Topics arising in the numerical solution of differential equations include

- non-stiff problems and stiff solvers;
- geometric solvers;
- parameter estimation and Bayesian approaches, where the vector field may depend on a model parameter θ , as in $f(t, x, \theta)$;
- boundary-value problems; and
- stochastic differential equations (SDEs).

We generally consider vector-valued solutions and vector fields,

$$x: \mathbb{R} \longrightarrow \mathbb{R}^n, \quad f: \mathbb{R} \times \mathbb{R}^n \longrightarrow \mathbb{R}^n,$$

which can be written componentwise as

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad f(t, x) = \begin{bmatrix} f_1(t, x_1, \dots, x_n) \\ \vdots \\ f_n(t, x_1, \dots, x_n) \end{bmatrix}.$$

1.2. Examples.

Example 1. *The scalar equation*

$$x' = x$$

is an ODE with $n = 1$.

Example 2 (Damped oscillator). *Consider the second-order equation*

$$x'' = -\omega^2 x - x'.$$

Setting $x_1 = x$ and $x_2 = x'$ converts it into the first-order system

$$\begin{cases} x'_1 = x_2, \\ x'_2 = -\omega^2 x_1 - x_2. \end{cases}$$

Thus, with $\mathbf{x} = (x_1, x_2)^\top$, we have $\mathbf{x}' = \mathbf{f}(x_1, x_2)$.

Example 3 (Chemical kinetics). *A three-species kinetic model is*

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \begin{cases} x'_1 = -k_1 x_1 + k_2 x_2 x_3, \\ x'_2 = k_1 x_1 - k_2 x_2 x_3 - k_3 x_2^2, \\ x'_3 = k_3 x_2^2, \end{cases}$$

where $k_1, k_2, k_3 > 0$ are reaction rates. The reaction rates may occur on very different scales, which can lead to stiffness.

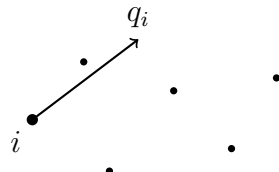
Example 4 (Molecular model). *Let q_i be the position of particle i , let m_i be its mass, and let $V(q_1, \dots, q_N)$ be the potential energy. Newton's equation is*

$$m_i q_i'' = -\frac{\partial V(q_1, \dots, q_N)}{\partial q_i}.$$

Introducing the momentum $p_i = m_i q'_i$ gives

$$\begin{cases} q'_i = \frac{p_i}{m_i}, \\ p'_i = -\frac{\partial V}{\partial q_i}. \end{cases}$$

The positions and momenta can therefore be collected into the state vector

$$\mathbf{x} = \begin{bmatrix} q_1 \\ p_1 \\ q_2 \\ p_2 \\ \vdots \\ q_N \\ p_N \end{bmatrix}.$$


1.3. Solutions and the Lipschitz condition. Consider the IVP (1) on an interval containing t_0 .

Definition 1. A function $x(t)$ is a solution of the IVP if it satisfies the integral equation

$$x(t) = x_0 + \int_{t_0}^t f(s, x(s)) ds.$$

Definition 2 (Lipschitz condition). The vector field f is Lipschitz continuous in x on a domain D if there is a constant $L \geq 0$ such that

$$\|f(t, x) - f(t, y)\| \leq L\|x - y\|$$

for all $x, y \in D$ and all relevant t .

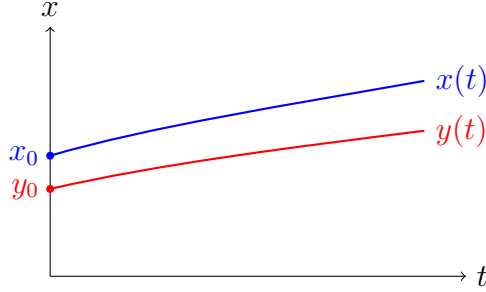
Under the usual continuity assumption in t , the Lipschitz condition in x implies that the solution of the IVP exists and is unique on an appropriate interval.

The same condition also controls the sensitivity to the initial data. Let $x(t)$ and $y(t)$ be solutions with initial values x_0 and y_0 , respectively. Their integral equations give

$$\begin{aligned}\|x(t) - y(t)\| &\leq \|x_0 - y_0\| + \int_{t_0}^t \|f(s, x(s)) - f(s, y(s))\| ds \\ &\leq \|x_0 - y_0\| + L \int_{t_0}^t \|x(s) - y(s)\| ds.\end{aligned}$$

By Grönwall's inequality,

$$(2) \quad \|x(t) - y(t)\| \leq \|x_0 - y_0\| e^{L(t-t_0)}.$$



Thus nearby initial values produce solutions whose separation is bounded by (2).